OmegaSim/OmegaHive Strange-Attractor Tuning: A Hyperseed-Dynamical Formalization

OmegaClaw / ZeroBot working notes for Ben Goertzel

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Abstract

This note formalizes a useful mathematical frame for constructing and tuning OmegaSim/OmegaHive-style simulations so that they can explore complex bounded nonlinear dynamics, including operational strange-attractor regimes. The core idea is to model hives as delayed, coupled, dissipative nonlinear systems; to express tuning through dimensionless control ratios; to prove what follows from well-posedness, boundedness, contraction, and delay-instability arguments; and to translate the resulting theory into concrete simulation recommendations. The claims here are deliberately split into theorems, conjectures, operational criteria, and implementation suggestions.

1 Provenance and aim

This note responds to Ben Goertzel's 2026-06-28 request for a rigorous mathematical formalization of the "useful frame" proposed for tuning one-hive and OmegaHive-style simulations toward complex strange-attractor dynamics. The requested steps are:

1. formalize concepts bridging Hyperseed and nonlinear dynamics;
2. state and prove general-purpose theorems applicable across agent hives;
3. specialize these results to OmegaHive1/OmegaSim-style simulations;
4. give theory-motivated instructions for the Codex/OmegaSim simulation agent.

The purpose is not to prove that a given current simulation has a strange attractor. Rather, it is to identify mathematically meaningful design knobs and verification gates so that such a claim can later become evidence-bearing.

2 S1. Concepts bridging Hyperseed and nonlinear dynamics

2.1 Hyperseed event schema as a state observable

Definition 1 (Hyperseed process event)**.** *A Hyperseed-relevant process event is a typed record*

$$e = \langle \iota, \tau, A, C, R, P, \Theta, \nu \rangle$$

where ι is identity, τ time, A the active agents or sub-processes, C context, R realized relation or result, P provenance, Θ an interpretive type assignment, and ν uncertainty/truth-value metadata.

Definition 2 (Hive observable)**.** *A hive observable is a map*

$$O : E_{\le t} \to X \subseteq \mathbb{R}^n$$

from a finite or filtered history of Hyperseed process events to a numerical state vector. Components may include agent activation, resource load, attention, coordination phase, memory pressure, trust, artifact quality, task backlog, message volume, or semantic diversity.

Definition 3 (Scrutable reduction)**.** *A reduction from Hyperseed events to dynamical state is scrutable when each coordinate x_i of $x = O(E)$ has: (i) a declared meaning, (ii) a measurement rule, (iii) provenance to events or logs, and (iv) an uncertainty model. Unscrutable coordinates may be useful for machine learning, but should not carry ontological interpretation without an additional explanation layer.*

2.2 Delayed agent-hive dynamics

Let $x(t) \in X \subseteq \mathbb{R}^n$ be a numerical hive state derived from the observable map above. Let the delayed history segment be

$$x_t(\theta) = x(t+\theta), \qquad \theta \in [-\tau, 0],$$

where $\tau \ge 0$ is a maximum communication, reaction, or memory delay.

Definition 4 (Continuous delayed hive system)**.** *A continuous delayed hive system is a retarded functional differential equation*

$$\dot{x}(t) = F(x_t, u(t), \eta(t); \theta),$$

where $F : C([-\tau, 0], X) \times U \times E \to \mathbb{R}^n$, $u(t)$ is a controllable intervention/tuning signal, $\eta(t)$ is bounded environmental or stochastic forcing, and θ denotes fixed parameters.

Definition 5 (Discrete delayed hive simulation)**.** *A discrete delayed hive simulation is a lifted map*

$$x_{k+1} = G(x_k, x_{k-1}, \dots, x_{k-m}, u_k, \eta_k; \theta),$$

with lifted state

$$z_k = (x_k, x_{k-1}, \dots, x_{k-m}) \in X^{m+1}.$$

Equivalently,

$$z_{k+1} = \widehat{G}(z_k, u_k, \eta_k; \theta) = (G(x_k, \dots, x_{k-m}, u_k, \eta_k), x_k, \dots, x_{k-m+1}).$$

2.3 Dimensionless tuning coordinates

The raw constants of a hive simulation are usually too numerous to sweep intelligently. We therefore define dimensionless ratios.

Definition 6 (Dimensionless hive control ratios)**.** *For a delayed nonlinear hive model, define:*

$$\rho = g\, r(A), \qquad \delta = \frac{\tau}{T_{\text{relax}}}, \qquad \mu = \frac{T_{\text{memory}}}{T_{\text{decision}}},$$

$$\kappa = \frac{\|A_{\text{cross}}\|}{\|A_{\text{within}}\|}, \qquad \nu = \frac{\sigma_\eta}{\|F_0\| + \epsilon},$$

where g is a typical nonlinear response gain, $r(A)$ is the spectral radius of an interaction matrix or linearized coupling operator, T_{relax} is local relaxation time, T_{memory} memory decay time, T_{decision} update time, A_{cross} and A_{within} are cross-hive and within-hive couplings, and σ_η is noise magnitude.

Interpretation: ρ measures feedback amplification, δ delay pressure, μ persistence, κ modular leakage, and ν noise-to-drift. These are not universal constants; they are organizing coordinates for phase diagrams.

Definition 7 (Operational strange-attractor regime)**.** *A parameter region $\mathcal{P}$ is an operational strange-attractor regime for a simulation if, for a specified observable O and lifted state z, the following hold robustly over validation runs:*

1. *trajectories are ultimately bounded and not merely saturating at hard numerical limits;*
2. *the largest finite-time Lyapunov estimate is positive under controlled perturbations of initial histories;*
3. *long-run statistics are stable under timestep/refinement and seed checks;*
4. *recurrence plots, broadband spectra, or reconstructed dimensions indicate non-periodic structured recurrence rather than white noise.*

This is an empirical criterion. It is weaker than a proof of a mathematical strange attractor.

3 S2. General theorems

3.1 Well-posedness and boundedness

Theorem 1 (Local well-posedness for delayed hives)**.** *Assume F is locally Lipschitz in the history variable x_t, uniformly over bounded u and η. Then for every continuous initial history $\phi \in C([-\tau, 0], X)$ there exists a unique maximal solution.*

Proof. This is the Picard–Lindelof theorem for retarded functional differential equations. The history space $C([-\tau, 0], X)$ is a Banach space under the sup norm. Local Lipschitz continuity gives a contraction on a sufficiently short interval. Repetition extends the solution until it exits every bounded subset of the state space. □

Theorem 2 (Ultimate boundedness from dissipativity)**.** *Suppose there is a radially unbounded C^1 function $V : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ and constants $a, b > 0$ such that along all trajectories with bounded input magnitude M,*

$$\dot{V}(x(t)) \leq -aV(x(t)) + bM.$$

Then

$$\limsup_{t\to\infty} V(x(t)) \leq \frac{bM}{a}.$$

Proof. By Gronwall's inequality,

$$V(t) \leq e^{-at}V(0) + \frac{bM}{a}(1 - e^{-at}).$$

Taking $t \to \infty$ gives the result. □

Corollary 1 (Attractor-talk requires a bounded regime)**.** *A simulation run that diverges, explodes numerically, or only bounces against uninterpreted clipping bounds does not by itself justify talk of an attractor. Dissipativity or empirically verified boundedness is a prerequisite diagnostic.*

Theorem 3 (Forward-invariant box for delayed maps). *For*

$$x_{k+1} = G(x_k, \dots, x_{k-m}, u_k),$$

assume there is a compact set $K \subseteq \mathbb{R}^n$ such that

$$G(K^{m+1}, U) \subseteq K.$$

Then K^{m+1} is forward invariant under the lifted map $\widehat{G}$.

Proof. If $z_k = (x_k, \dots, x_{k-m}) \in K^{m+1}$, then each delayed component lies in K. By assumption the new component $x_{k+1} = G(\cdots)$ lies in K. The remaining components of z_{k+1} are shifted copies of previous components in K. Therefore $z_{k+1} \in K^{m+1}$. $\square$

3.2 Contraction: when complex dynamics is impossible

Theorem 4 (Delayed-map contraction implies a unique attracting history). *Let $\widehat{G} : K^{m+1} \to K^{m+1}$ be the lifted delayed map on compact convex K^{m+1}, and suppose it is a contraction with Lipschitz constant $L < 1$ under some norm. Then there is a unique fixed lifted state z^*, and every trajectory converges to z^*.*

Proof. This is Banach's fixed-point theorem applied to $\widehat{G}$ on the complete metric space K^{m+1}. Iterates of a contraction converge to the unique fixed point. $\square$

Corollary 2 (Low-gain hives cannot be chaotic). *If the effective lifted update of a hive simulation is contractive, then it cannot have a strange attractor, quasiperiodic orbit, or sustained multi-cycle behavior. Therefore one route toward complex dynamics is to increase gain, delay, cross-coupling, or memory persistence until the lifted system leaves the contraction regime while preserving boundedness.*

Proposition 1 (A practical Lipschitz bound). *Suppose*

$$G(x_k, \dots, x_{k-m}) = S\left(B + \sum_{j=0}^{m} A_j x_{k-j}\right)$$

where S is componentwise Lipschitz with constant s and the lifted norm is chosen so that the shift has unit weight. A sufficient condition for contraction is that the induced lifted Lipschitz constant of the companion map is less than one; in particular the response gain s and delayed coupling norms $\|A_j\|$ provide an upper bound. Thus the product "gain times coupling" is the first order control axis.

Proof. The non-shift component changes by at most $s \sum_j \|A_j\| \, \|\Delta x_{k-j}\|$ under the chosen induced norm, while the shift components copy previous differences. Weighted norms can penalize older delay coordinates so that the whole companion map is contractive when the response and coupling norms are sufficiently small. This proves the sufficient, not necessary, condition. $\square$

3.3 Delay-induced oscillatory instability

Theorem 5 (Scalar delay Hopf boundary). *Consider*

$$\dot{x}(t) = -ax(t) + bx(t - \tau), \qquad a > 0.$$

The characteristic equation is

$$\lambda + a - be^{-\lambda\tau} = 0.$$

A Hopf crossing candidate satisfies $\lambda = i\omega$, hence

$$a = b\cos(\omega\tau), \qquad \omega = -b\sin(\omega\tau).$$

Therefore delayed feedback can destabilize an equilibrium through an oscillatory root crossing even when the local undelayed term is dissipative.

Proof. Substitute $x(t) = e^{\lambda t}$ to obtain the characteristic equation. At an oscillatory stability boundary $\lambda = i\omega$. Separating real and imaginary parts gives the two displayed equations. □

Corollary 3 (Why $\delta = \tau/T_{\text{relax}}$ matters)**.** *The ratio of communication/memory delay to local relaxation time is a genuine bifurcation coordinate, not just an implementation detail. Increasing delay can convert damped consensus into oscillation and, in higher-dimensional nonlinear systems, into period-doubling or more complex regimes.*

3.4 Existence of compact attractors

Theorem 6 (Compact global attractor under dissipative smoothing)**.** *For a continuous delayed hive semiflow on $C([-\tau, 0], X)$, assume: (i) unique forward solutions exist, (ii) solutions enter a bounded absorbing set, and (iii) the semiflow is asymptotically compact. Then there is a compact global attractor $\mathcal{A}$ attracting all bounded sets.*

Proof. This is the standard global-attractor theorem for continuous semiflows: a bounded absorbing set plus asymptotic compactness implies the omega-limit of the absorbing set is nonempty, compact, invariant, and attracts every bounded set. □

Corollary 4 (Delay makes rigor harder)**.** *Delayed systems are infinite-dimensional in history space. Compactness and low fractal dimension are not automatic. OmegaSim should therefore treat finite state-history simulations as approximations and validate attractor claims under history-length and timestep variation.*

3.5 What remains conjectural

Conjecture 1 (Useful complex regime window)**.** *For many OmegaHive-like systems there is a practical window between contraction and blow-up/saturation, characterized roughly by $\rho \approx 1$, non-negligible δ, moderate μ, and controlled κ, in which bounded complex recurrence is most likely.*

This is not a theorem because it depends on nonlinear form, topology, stochastic forcing, and measurement. It is a theory-guided search heuristic.

Conjecture 2 (Hyperseed-scrutable complexity)**.** *The most scientifically useful OmegaSim regimes will be those where the low-dimensional observables show complex recurrence while Hyperseed event records remain interpretable enough to explain which agent roles, memories, artifacts, and feedback loops generate the recurrence.*

4 S3. Specialization to OmegaHive1 / OmegaSim

Since the exact OmegaHive1 simulation state is still evolving, define a generic OmegaHive1 state vector as

$$x_k = (a_k, r_k, m_k, q_k, h_k, c_k, p_k) \in \mathbb{R}^n,$$

where a is agent activation/attention, r resource or compute load, m memory persistence/pressure, q artifact quality or progress, h semantic diversity, c coordination/coherence, and p task pressure. The components may be vectors indexed by agents, roles, artifacts, or sub-hives.

A useful discrete OmegaHive1 skeleton is

$$x_{k+1} = S\left(W_0 x_k + W_1 x_{k-d} + W_m M_k + B u_k + \xi_k\right) - D x_k,$$

with memory update

$$M_{k+1} = (1-\alpha) M_k + \alpha H(x_k, \mathrm{events}_k),$$

where S is saturating nonlinear response, W_0 present-time coupling, W_1 delayed coupling, M_k memory state, u_k interventions, ξ_k controlled noise, and D local decay.

Proposition 2 (OmegaHive1 boundedness by saturation and resource accounting)**.** *If each component of S maps into a compact interval K_i and resource/memory updates are either projected into compact budgets or dissipative in the sense of the boundedness theorem, then the lifted OmegaHive1 history has a compact forward invariant set.*

Proof. The saturating components satisfy the invariant-box theorem. Resource and memory coordinates either remain in explicit compact budgets or satisfy dissipative bounds. The product of these compact coordinate sets is compact and invariant under the lifted delayed map. □

Proposition 3 (OmegaHive1 contraction baseline)**.** *If the fitted or designed Lipschitz constant of the lifted OmegaHive1 update is below one, the system will converge to a unique stable regime and should not be expected to exhibit strange-attractor-like dynamics.*

Proof. Immediate from the delayed-map contraction theorem. □

Proposition 4 (OmegaHive1 delay/gain destabilization target)**.** *Let x^* be an equilibrium of a smooth deterministic OmegaHive1 skeleton. Linearizing gives*

$$\delta x_{k+1} = J_0 \delta x_k + J_1 \delta x_{k-d} + J_m \delta M_k.$$

The stability of x^* is determined by the eigenvalues of the lifted companion Jacobian. Parameter sweeps that increase response gain, delayed coupling, or memory persistence move these eigenvalues toward or beyond the unit circle.

Proof. Linearization of the lifted map gives an ordinary finite-dimensional map on history and memory coordinates. Local discrete-time stability is equivalent to all eigenvalues of the companion Jacobian lying inside the unit disk. Increasing the norms of coupling and persistence terms changes this Jacobian continuously; instability occurs when eigenvalues cross the unit circle. □

4.1 OmegaHive1 order parameters

For diagnosis, define low-dimensional order parameters:

$$R_k = \left| \frac{1}{N} \sum_{j=1}^{N} e^{i\phi_{j,k}} \right| \quad \text{(coordination/synchrony)},$$

$$H_k = -\sum_{a} p_k(a) \log p_k(a) \quad \text{(behavioral or semantic diversity)},$$

$$C_k = x_k^\top L x_k \quad \text{(network tension across graph Laplacian } L\text{)},$$

$$Q_k = \text{artifact-progress or quality metric}, \qquad B_k = \text{resource/backlog pressure}.$$

Analyze trajectories in (R, H, C, Q, B) as well as in raw state space.

Design Principle 1 (Do not search for chaos in raw logs first)**.** *Raw agent traces are too high-dimensional and semantically mixed. First define scrutable observables and order parameters, then test whether the induced observable dynamics has bounded recurrence, sensitivity, and nonperiodicity.*

5 S4. Theory-motivated OmegaSim instructions

5.1 Implementation targets

1. Implement the lifted delayed state $z_k = (x_k, \ldots, x_{k-m}, M_k)$ explicitly. Do not hide delays in ad hoc buffers without logging them.
2. Define a compact state budget. Use saturating nonlinearities, normalization, or explicit resource conservation so runs cannot be mistaken for attractors while diverging.
3. Expose the control ratios $\rho, \delta, \mu, \kappa, \nu$ as primary sweep parameters. Raw constants should be derived from these ratios where possible.
4. Compute the fixed point or empirical baseline regime at low gain. Use it as the contraction/control case.
5. Linearize the deterministic skeleton around equilibria or recurrent regimes. Track eigenvalues of the lifted companion Jacobian as parameters change.
6. Run sweeps that cross from $\rho < 1$ to $\rho > 1$, from $\delta \approx 0$ to $\delta = O(1)$, and from weak to moderate memory persistence μ.
7. Near transitions, run hysteresis sweeps upward and downward in gain/delay to detect multistability and crises.
8. Estimate finite-time Lyapunov exponents with paired trajectories, periodic renormalization, multiple perturbation directions, and matched noise paths.
9. Add null controls: no-delay, linearized nonlinearity, degree-preserving rewiring, delay-shuffled, phase-randomized surrogate, and frozen-noise paired comparisons.
10. Classify each region as fixed point, limit cycle, quasiperiodic, candidate chaotic, transient chaotic, noise-driven irregular, divergent, or trivial saturation.

5.2 Specific suggested sweep grid

A first OmegaSim phase diagram should sweep:

$$\rho \in [0.2, 3.0], \qquad \delta \in \{0, 0.1, 0.3, 1, 3\},$$

$$\mu \in \{0.5, 1, 3, 10\}, \qquad \kappa \in \{0, 0.05, 0.2, 0.8\}, \qquad \nu \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}.$$

The exact ranges should be rescaled after measuring the relaxation time and state norms of the implemented simulator.

5.3 Acceptance criteria for a candidate strange-attractor region

A region should only be reported as a candidate complex/strange-attractor regime if it passes all of these gates:

1. boundedness gate: norms and resource variables remain in a nontrivial compact range;
2. refinement gate: the qualitative regime survives smaller timestep or longer history resolution;
3. perturbation gate: positive finite-time Lyapunov estimate under fixed noise path;
4. surrogate gate: behavior differs from no-delay, linearized, and phase-randomized controls;
5. semantic gate: Hyperseed event records identify plausible feedback loops generating the dynamics;
6. task gate: the regime is related to useful simulation behavior, not just aesthetic irregularity.

6 Conclusion

The rigorous core is modest but useful. If the hive update is ill-posed, unbounded, or contractive, complex attractor claims are unsupported or impossible. If the system is well-posed, bounded, non-contractive, delayed, and nonlinear, then gain, delay, memory persistence, modular leakage, and noise become principled coordinates for exploring bifurcations and candidate strange-attractor regimes.

The practical OmegaSim instruction is therefore: build the simulator so these coordinates are explicit; enforce boundedness; sweep through the contraction boundary; test delay-induced instability; and only call a regime strange-attractor-like after Lyapunov, recurrence, surrogate, refinement, and semantic-provenance checks pass.